

Continuous Distillation

Pre-Lab Plan Unit Operations Lab #6

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Group 2, Tuesday, PM section

Lindsey Meisch

Kendall Johnson

Christian Salinas Rojas

Srija Bomma

1. Experimental Objective

The objective of the experiment is to investigate the steady-state behavior of a continuous distillation column to separate an ethanol-water mixture and to determine how operating conditions affect the performance of the separation and the productivity of the column.

2. Experimental

An ethanol-water mixture under steady-state conditions was used in a continuous distillation column. A reboiler was loaded with a 10 wt% ethanol solution, and a 20 wt% ethanol solution was used to feed. The column was first run at total reflux to enable the system to stabilize at a steady state as shown by the stabilized tray temperatures.

A steady state was obtained and then the system was switched to continuous operation using a reflux ratio of about 5:1. At the column center, the feed was added at a fixed flow rate, and the distillate and bottoms of streams were collected.

Samples of the top and bottom products were taken at regular time intervals and analyzed using a densitometer to determine ethanol composition. Tray temperatures were also taken to observe column performance and to proceed to a steady state. A McCabe-Thiele analysis was performed on the experimental data to identify the number of theoretical stages, as well as to compare experimental data with ideal equilibrium behavior to analyze column efficiency.

2.1 Apparatus (5 pts)

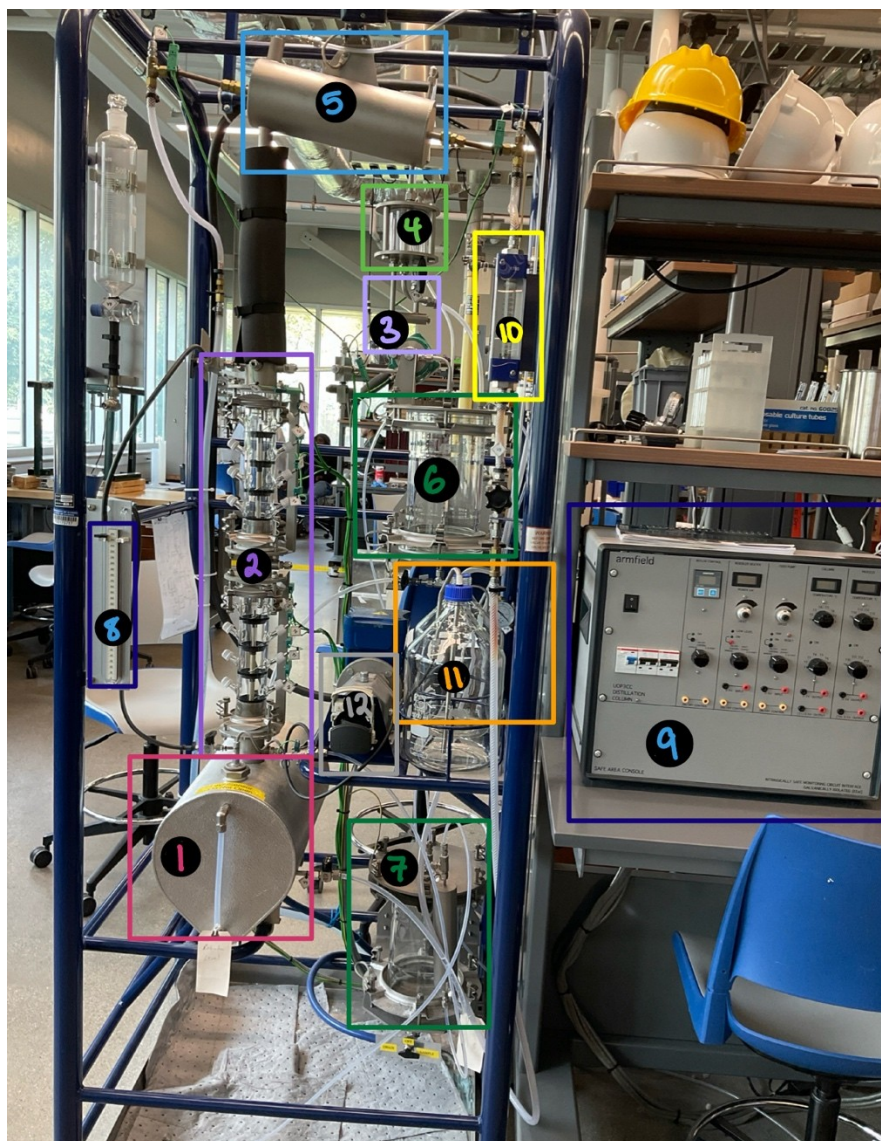


Figure 1: Apparatus with color coded and numbered labeling.

Table 1: Apparatus parts correlating to Figure 1.

Number/Color	Apparatus Part
1 – Pink	Reboiler w/ heater
2 – Purple	Distillation Column (with thermocouples)
3 – Light Purple	Reflux Valve
4 – Light Green	Decanter
5 – Light Blue	Condenser
6 – Dark Green	Top Product Receiver

7 – Dark Green	Bottom Product Receiver
8 – Dark Blue	Manometer
9 – Dark Blue	Safe Area Console
10 – Yellow	Cooling water manometer
11 – Orange	Feed drums
12 – Grey	Peristaltic pump

2.2 Materials and Supplies (5 pts)

Table 2: Materials and supplies used in experiment and their purpose.

Material/Supplies	Purpose/Specifications
200 Proof Ethanol	For the ethanol/water mixture that is being distilled
DI Water	For the ethanol/water mixture that is being distilled
Cooling Water	Used for the condenser
Funnel	To get the ethanol/water mixture into the reboiler
Graduated Cylinder	To make the ethanol/water mixture
Beaker	For various product collection, and making the ethanol/water mixture
Test Tube	For product collection, and testing
Parafilm	To avoid evaporation of hot product – to keep the ethanol/water composition from changing of the product as the product cools off
Portable Alcohol Reader (Densitometer)	To measure the composition of the product
Computer	To collect data from the Safe Area Console
Water Bath	Cooling down the samples
Test tube rack	For holding test tubes before and after taking samples; for the ice bath
Stir rods	For holding test tube rack within ice bath
Masking tape	Labeling the samples

2.3 Experimental Plan (10 pts)

- [Prelab] Calibrate the peristaltic feed pump. Disconnect the feed tubing from the column, place the suction tubing in water, and measure the delivered flow rate with a graduated cylinder at ten settings in timed intervals across the dial range (1-10). Use these measurements to construct a calibration curve of flow rate vs dial position so the desired feed rate can be selected during the experiments.
- Prepare the working solutions for the column: 10L of 10 wt% ethanol in water for the reboiler and 3 L of 20 wt% ethanol in water for the feed tank. Make sure the mixtures are prepared carefully so that the starting compositions are known as accurately as possible.
- Charge the reboiler with the 10 wt% ethanol solution and fill one feed bottle with the 20 wt% ethanol solution. Confirm that the tubing is connected correctly and that the feed line is initially attached to the midpoint feed location between trays 4 and 5 for the first steady-state experiment.
- Check that the valves are in the proper starting positions, open the reflux line as required, and begin condenser cooling water flow at approximately 3 L/min. Then turn on the control panel and energize the reboiler. Initially heat the reboiler to about 1.5 kW to bring the column up to temperature, then reduce the power to about 0.75 kW once vapor flow is established.
- Operate the column first at a total reflux until the tray and reboiler temperatures stabilize, indicating that the column is approaching steady operation. Monitor temperatures T1 through T9 during this warm up period to observe how the temperature profile develops along the column.
- Once the system has stabilized, begin the first steady state run by setting the reflux ratio to 5:1, opening the bottoms valve, and starting the feed pump. Use the pump calibration curve to set the feed flow to approximately 30-35 mL/min. This first run will represent the conventional feed-at-midpoint configuration.
- After approximately 400 mL of feed has been introduced, collect one overhead sample and one bottoms sample. Then collect two additional sets of overhead and bottoms samples, spaced 5 minutes apart, while also recording the full column temperature profile (T1-T9) for each sampling period. Measure sample composition using the SNAP-41 densitometer.
- After the first run is complete, stop the feed pump, turn off the reflux switch, shut down the reboiler, and close the necessary valves. If needed, allow the system to cool slightly before changing the feed location.
- For the second experiment, disconnect the feed from the midpoint location and reconnect it to the feed point above the top tray, creating the stripping column configuration. Reheat the system to the same operating condition and again allow the column to stabilize under total reflux.
- Repeat the same operating sequence used in the first run: set the reflux ratio to 5:1, open the bottoms stream, start the feed at approximately 30-35 mL/min, allow about 400 mL of feed to enter, and then collect three overhead samples and three bottoms samples, each taken 5 minutes apart, while recording temperatures T1-T9. Measure sample compositions with the densitometer.

- Shut down the system by turning off the feed pump, reflux switch, and reboiler power, then close valves and leave the equipment in a safe condition.

The experiment consists of three main parts: feed pump calibration, steady state distillation run with feed introduced at the midpoint, and a steady state distillation run with feed introduced above the top tray. The calibration is completed first so that the feed flow during both distillation runs can be set more accurately. Both steady state experiments will be conducted at a reflux ratio of 5:1, with a nominal feed rate of 30-35 mL/min, a reboiler operating power of about 0.75 kW after startup, and condenser cooling water near 3 L/min. The order of the runs is important because the midpoint-feed case is the standard column operation, while the second run isolates the effect of the feed location by changing only that variable and holding the other operating conditions as constant as possible.

The main independent variable in this experiment is the feed position in the column, which is tested at two locations: between trays 4 and 5 and above the top tray. A second controlled operating variable is the feed flow rate, which is targeted at 30-35 mL/min using the calibration curve, while the reflux ratio is held constant at 5:1. The principal dependent variables are the ethanol composition of the distillate and bottoms products and the temperature profile along the column from T1 to T9. Product compositions are measured from liquid samples using the densitometer, and temperatures are recorded directly from the column instrumentation. These measured responses indicate how effectively the column separates ethanol and water under each feed configuration.

For each steady state operating condition, the plan includes three overhead samples and three bottoms samples, collected at 5-minute intervals. These serve as replicate measurements at steady state and can be used to calculate an average composition and associated variability for each stream. The mean will be reported as well as a 95% confidence interval for the measured top and bottom compositions based on the three replicate samples. Experimental uncertainty will be reduced by calibrating the feed pump before the distillation runs, using the same reflux ratio and power setting for both experiments, and sampling only after the column has stabilized. Additional uncertainty may arise from flow fluctuations, incomplete tray equilibrium, heat loss from the column, and measurement error in the densitometer and temperature readings.

The experimental data will be correlated with theory using an ideal McCabe-Thiele analysis for the ethanol-water system, assuming a saturated liquid feed as specified in the lab manual. The measured feed, distillate, and bottoms compositions, together with the imposed reflux ratio, will be used to construct the operating lines and estimate the theoretical number of stages for each feed configuration. These theoretical predictions will then be compared with the actual column behavior to estimate overall column efficiency. Goodness of fit can be assessed by comparing predicted and experimental compositions and by calculating a percent error or a simple average absolute deviation between theoretical and measured values. The measured tray temperature profile can also be compared qualitatively with expected equilibrium behavior to evaluate whether the column is truly at steady state and whether the assumptions of the ideal model are valid.

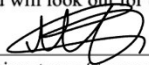
Project Safety Evaluation (10 pts)

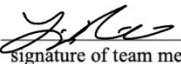
The primary chemical hazard in this experiment is ethanol, which is used in the ethanol–water mixtures circulating through the distillation column. Ethanol is flammable and can cause irritation or mild poisoning if significant exposure occurs through inhalation or ingestion. Proper laboratory practices should be followed when handling ethanol solutions, including working in a well ventilated space, avoiding spills, and wearing appropriate personal protective equipment such as safety glasses, gloves, and lab coats. Any spills should be cleaned immediately to prevent exposure and reduce fire hazards.

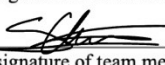
Thermal and mechanical hazards are also present due to the equipment used in the distillation system. The reboiler operates at elevated temperatures to generate vapor, meaning that surfaces on the column, tubing, and collected liquid samples may be very hot and can cause burns if handled improperly. Additionally, the peristaltic pump requires the pump head to be opened when inserting or adjusting tubing, which presents a minor pinch hazard if done while the system is operating. Care should also be taken to avoid spilling liquids near the nearby computer control system and electrical components. Ensuring equipment is handled carefully and allowing components to cool before touching them helps minimize these risks during operation of the distillation column.

	Yes/No	
Potential electrical hazards?	Y	If yes, identify conditions: Water by electrical components
Potential mechanical hazards?	Y	If yes, identify conditions: Peristaltic pump
Potential pressure hazards?	Y	If yes, identify conditions: Pressure release line
Potential temperature hazards?	Y	If yes, identify conditions: Reboiler (790°C), hot product
Hazardous/toxic chemicals involved?	Y	If yes, hazards involved: 200 proof ethanol
Gloves required?	Y	If yes, specify type: Heat resistant gloves, nitrile gloves
MSDS reviewed by all team members?	Y	List MSDS sheets reviewed: CAS # 64-17-5
Process Parameters	Max Expected	List location and possible hazards and precautions
Temperature (°C)	790°C	Generated by reboiler
Pressure (psia)	Atm	Should only deal w/ atm press
Safety Controls	yes/no	If yes, Provide description
	Y	heat & nitrile gloves, fume hood, L-L alarm in reboiler

I have read relevant background material for the Unit Operations Laboratory entitled: "Continuous Distillation" and understand the hazards associated with conducting this experiment. I have planned out my experimental work in accordance to standards and acceptable safety practices and will conduct all of my experimental work in a careful and safe manner. I will also be aware of my surroundings, my group members, and other lab students, and will look out for their safety as well.

 4/8/26
signature of team member date

 4/8/26
signature of team member date

 4/8/2026
signature of team member date

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Figure 2: Safety evaluation chart.